

Problem 6.21

You invest \$10,000 at time 0 into each of two accounts. Account A earns interest at an annual simple interest rate of 8%; Account B earns interest at an effective annual compound interest rate of 6%. What is the difference in the amount of interest earned during the 5th year in these two accounts?

Solution

For Account A with simple interest at 8%,

$$A_A(t) = 10000(1 + 0.08t).$$

Thus,

$$A_A(4) = 10000(1 + 0.32) = 13200$$

and

$$A_A(5) = 10000(1 + 0.40) = 14000.$$

So the interest earned during the 5th year is

$$I_5^{(A)} = 14000 - 13200 = 800.$$

For Account B with compound interest at 6%,

$$A_B(t) = 10000(1.06)^t.$$

Thus,

$$A_B(4) = 10000(1.06)^4 \approx 12624.7696$$

and

$$A_B(5) = 10000(1.06)^5 \approx 13382.2557.$$

So the interest earned during the 5th year is

$$I_5^{(B)} = 13382.2557 - 12624.7696 = 757.4861.$$

Therefore, the difference in the interest earned during the 5th year is

$$800 - 757.4861 = 42.5139.$$

42.5139